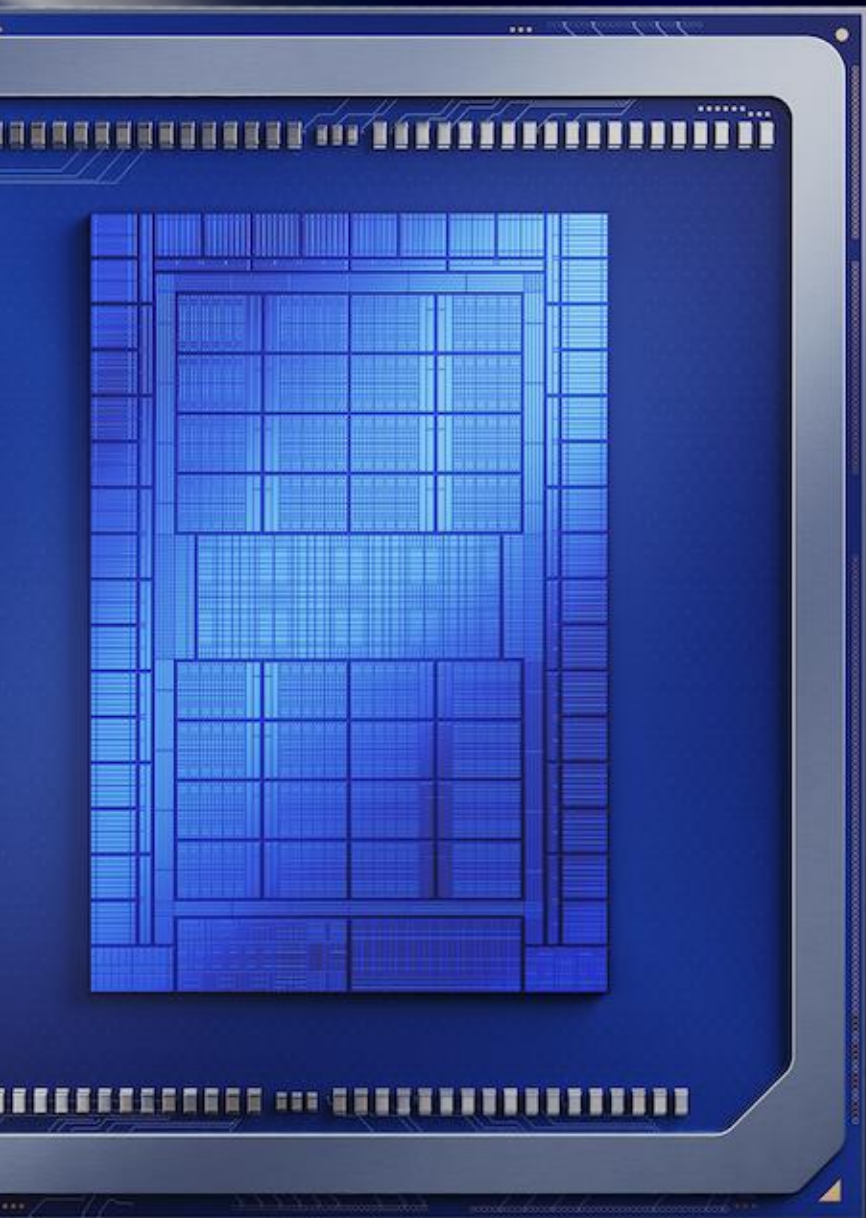




intel

Crescent Island: GPU Designed for Agentic AI Inference

Sumit Mohan, Chief Architect, Enterprise AI Systems

Dr. Hong Jiang, Intel Fellow, Chief GPU Compute Architect

Hot Chips – Aug 2026

Agentic AI Workload Compute Demands

Continuous Iteration



More Steps Drive
Latency Pressure

Single-turn inference time

Tool call turnaround time

Scheduling

Expanding Context



More Context Drives
Memory Pressure

Memory capacity

Long context handling

Memory Bandwidth

Concurrent Orchestration



More Agents Drive
System Throughput Pressure

CPU-GPU coordination

Heterogenous orchestration

Power density management

Accelerator Portfolio for Agentic AI

Scaling from local agents to large scale intelligence

Local Agent
Compute Boxes

Edge
AI

AI Builder
Workstations

Physical
AI

Enterprise
AI

Agentic AI
Data Centers

Local Agent Computers

Intelligence Centers

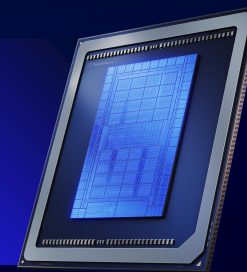
Integrated
**Intel® Arc™ Pro
GPUs**



Workstation
**Intel® Arc™ Pro
GPUs**



Data Center
**Intel®
GPUs**



 sambanova

**SN50
RDUs**



A Robust AI-forward GPU Architecture

Xe_{LP}

Low Power Graphics
for PC

11th Gen Intel® Core™

12th Gen Intel® Core™

2020

Xe

First Scalable Gfx & AI
for PC

Intel® Core™ Ultra Series 1

Intel® Core™ Ultra 200H series

Xe2

Efficiency First Gfx & AI
for PC & Workstation

Intel® Core™ Ultra Series 2

Intel® Arc™ Pro B-Series

Xe3

Gfx & AI Powerhouse
for PC & Edge

Intel® Core™ Ultra Series 3

TODAY'S TALK

Xe3P

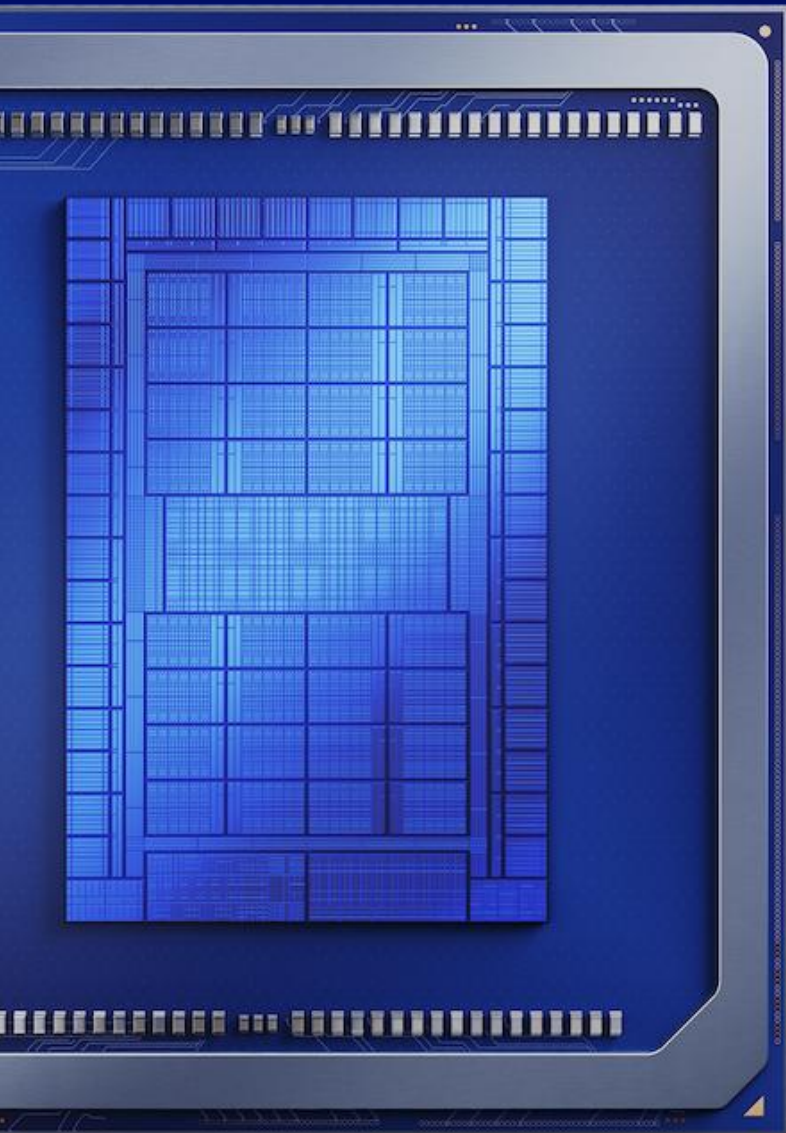
Performance-Forward
for PC, DC, Edge & Workstation

Crescent Island

Next Gen PC

...

2026+



Next Gen

Intel® GPU

Codenamed
Crescent Island

Designed for tokens/watt, built on a reliable open software stack

Built for Agentic AI

Xe3P

AI optimized GPU IP

FP4/MXFP4 - FP64

Widest range datatypes

Large Context Capability

up to 480GB

Capacity

LPDDR5x

Low power memory

Power Optimized

350W

Air-cooled PCIe

Open

& Robust software stack

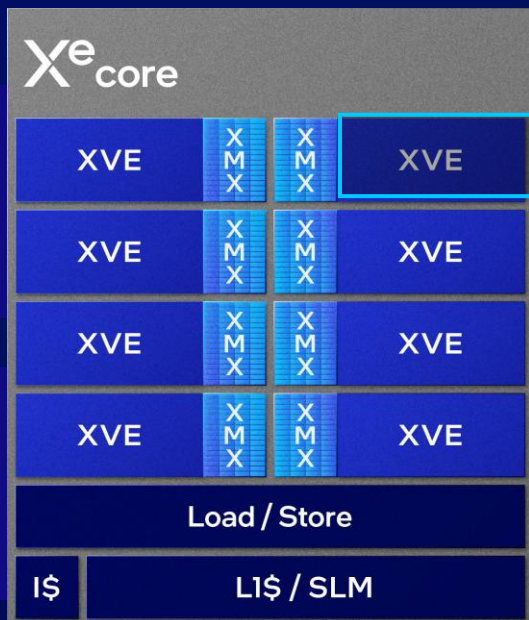
*Intel branded PCIe card has 160GB LPDDR5x; design enables partners to build ODM branded cards with flexible options up to 480GB memory.

3rd Generation

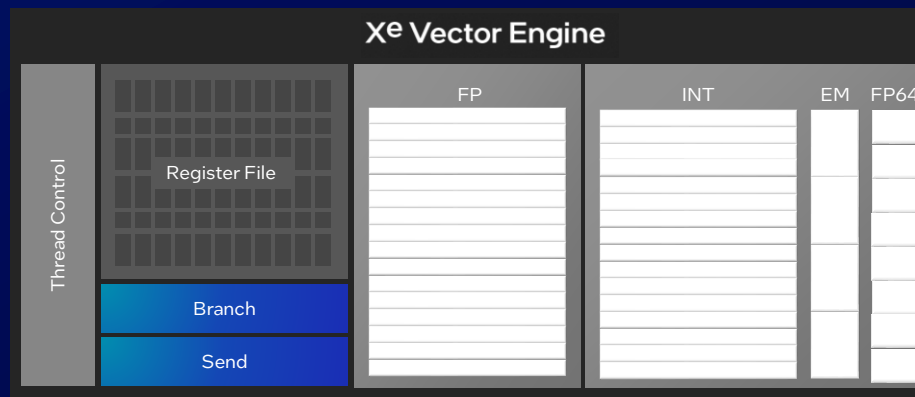
Xe core

8 Vector
Engines

8 XMX
Engines



Xe Vector
Engine



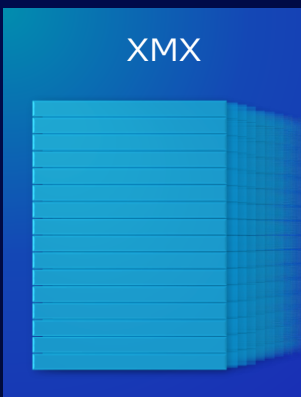
GPU IP with
FP4 Precision

3-way
co-issue

Extended
Math & FP64

Xe Matrix
Extensions

XMX



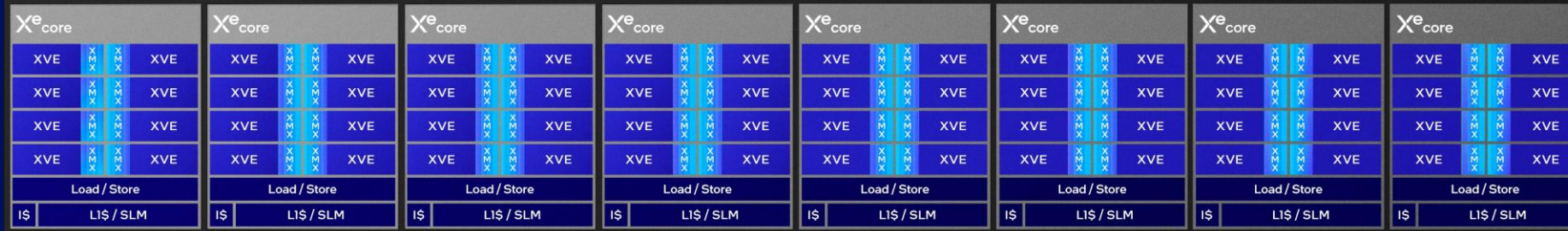
Xe Matrix
Extensions

AI Acceleration
Engines with 16-deep
systolic array

Optimized from engine to core for a more efficient performance

Crescent Island Architecture

Compute Slice



32 Xe-cores

256 XM engines

Crescent Island

Global Dispatch

Compute Slice



Memory Fabric

L2\$



Compute Slice



X^e-Architecture: Key Compute/AI Features over generations

X^e2

Codename: Battlemage Client dGPU (Arc B-series)

GPGPU
(FE, XeCore/EU)

20 Xe Cores
XMX: 4-deep systolic
FP16/BF16/TF32, INT8/INT4

FP64 (8 FMA/XeCore)

GRF per XeCore: 512KB

Memory
Sub-system

256KB L1\$/SLM per XeCore
18MB L2 cache
24GB GDDR6

X^e3

Codename: Panther Lake Client iGPU design

Up to 12 Xe Cores
XMX: 4-deep systolic
FP16/BF16/TF32, INT8/INT4

FP64 (8 FMA/XeCore)
Variable Registers per Thread (VRT)
Improved thread dispatch & preemption

GRF per XeCore: 512KB

Larger L1\$/SLM per XeCore

Shared system memory

X^e3p

Codename: Crescent Island AI GPU

32 XeCores
XMX: **16-deep systolic**
FP16/BF16/TF32, INT8/INT4
FP8 + FP4 format support
Microscaling (MX) format support

Full rate FP64 (**64 FMA**/XeCore)
Same Address Multi-Queue
Support for **Sigmoid and Tanh**
Transcendental/EM improvements

GRF per XeCore: **1MB**

512KB L1\$/SLM per XeCore
32MB unified L2
Up to 480GB LPDDR5x

Crescent Island SoC Chip



32 Xe Cores

High-FLOP
Compute Behind
Every Reasoning Step



32MB Unified L2

Bandwidth Filter
that Keeps the
Matrix Engines Fed



LPDDR5x

KV-Cache Capacity
for Long Context and
Concurrent Sessions



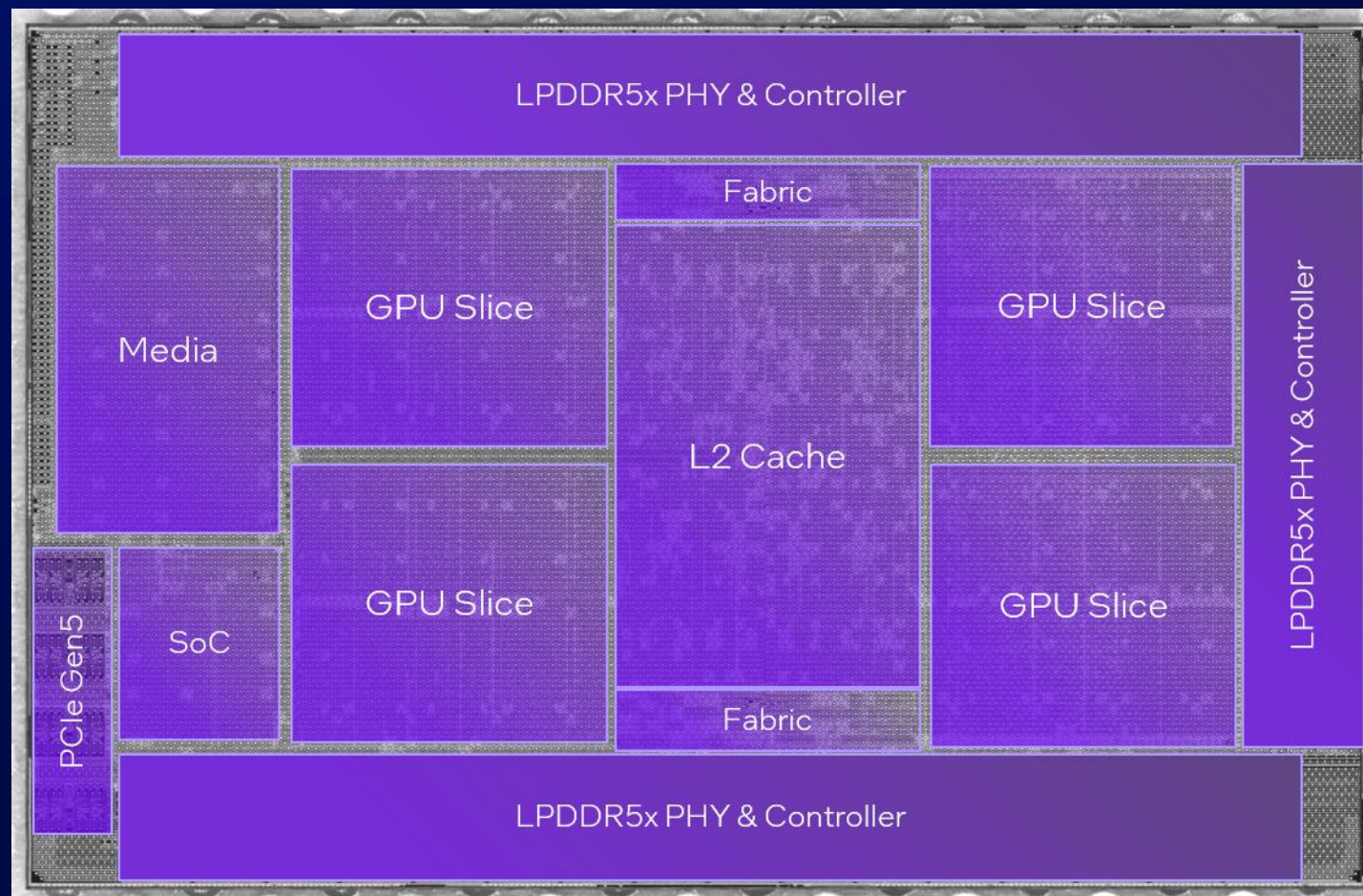
Media Engine

(4 Decoder + 4 Encoder)
Compressed-Domain
Visual I/O for
Multimodal Agents



PCIe Gen5 x16

Scale-Up on an
Open Standard
PCIe switch fabric



RAS Features



GPU IP and Media Reliability

ECC and Parity protection across key memory
and compute blocks

Fine grained error management such as masking
and thresholding

Local error containment and fault isolation at
Xe^ocore and memory subsystem level recovery

Dynamic Error Injection and Control

End-to-End Cmd/Address/Data Poison Propagation



LPDDR Memory Reliability

ECC over DMI enables drastically reduced SDC
rates without bandwidth / capacity impact

Dynamic Page Offlining

Dynamic Error Injection and Control

hPPR (hard Post Package Repair)

Data Poison Propagation

Patrol Scrubbing



PCIe Advanced Error Reporting (AER) support

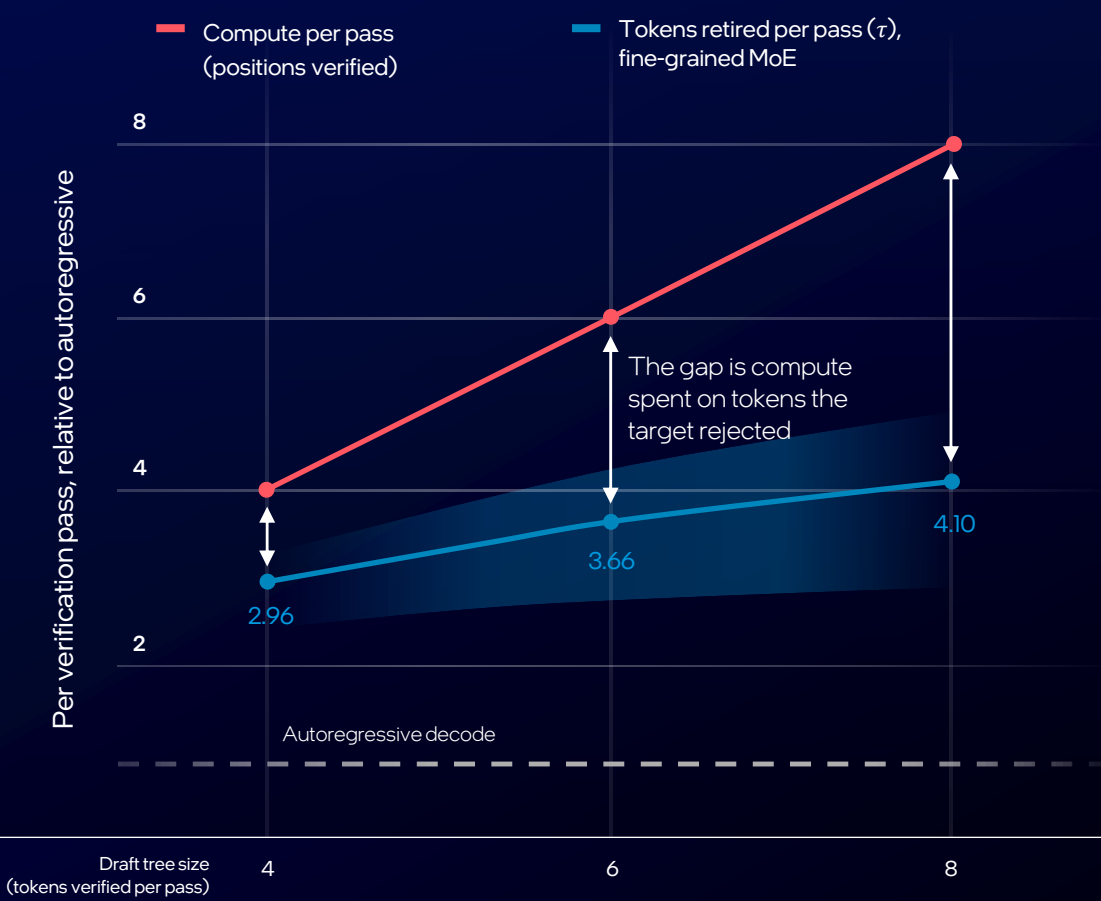
Decode Becomes a Compute Problem

Measured on frontier MoE: 2.9 to 4.9 tokens retired per verification pass, paid for in compute that autoregressive decode leaves idle

Published acceptance by routing class, 8-token draft tree



Doubling the tree doubles compute, buys 1.39x token

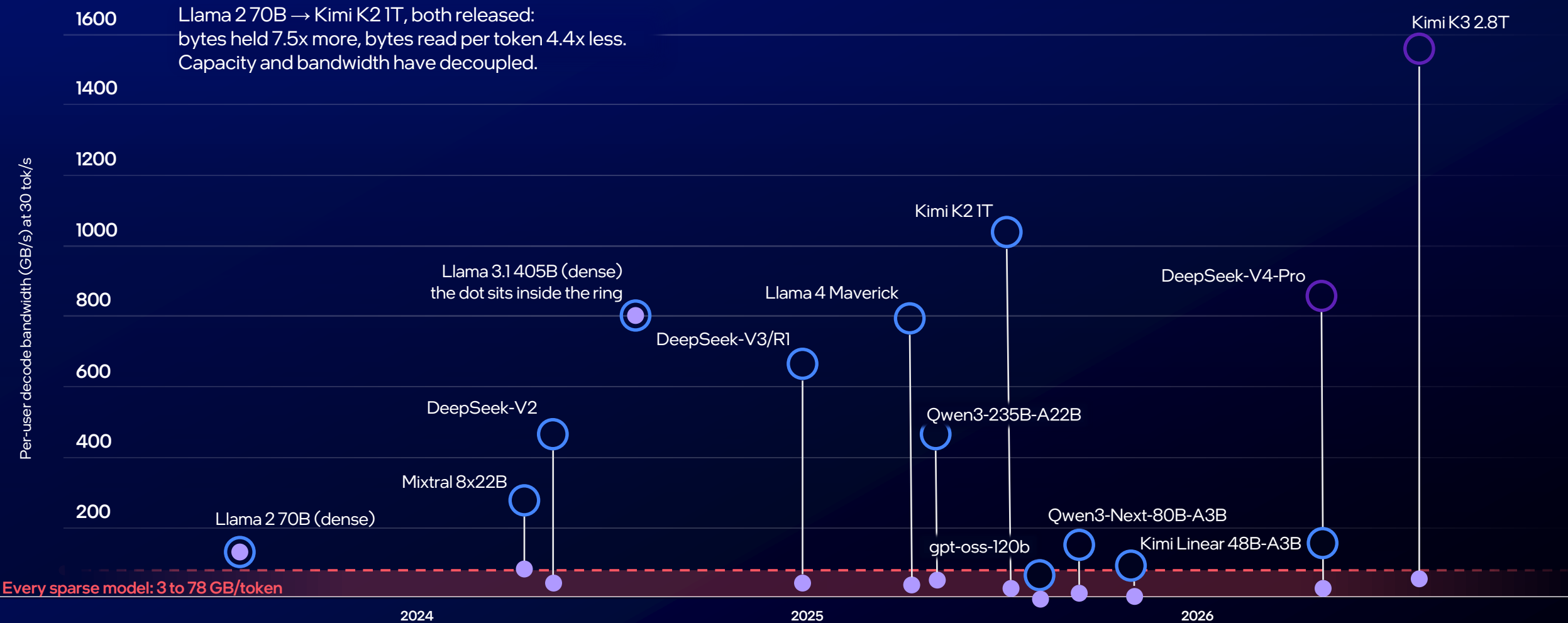


Source: LMSYS SpecBundle (SpecForge/SGLang) datasets, technical papers, and project announcements; [see appendix/reference slide for complete citations](#). Accepted lengths shown; throughput metrics excluded.

Bytes Held Grew 7x, Bytes Read Per Token Fell 4x

From published configs, Llama 2 70B to Kimi K2 1T; PCIe Gen5 switch scale-up grows capacity to hold them

- Total footprint: all weights resident (capacity)
- Active parameters: weights read per token (bandwidth)
- Announced; config not yet published



Benefits of higher memory capacity on Crescent Island

Run more concurrent sessions, long context workloads, and larger models for better TCO

More KV sessions

On same model size results in
higher aggregate throughput

Run long context RAG/agentic sessions

On same model size

Run larger models with fewer GPUs

Leading to better performance

**CRI can easily fit model weights and KV cache stored in FP8
while a GPU with lower memory cannot**

Crescent Island: designed for tokens/watt & agentic AI

Silicon Features

Agentic AI ready

High memory capacity and sustained compute

LPDDR5x memory

Higher density, lower power. Enhanced reliability with no BW/capacity loss

Performance optimized for AI

Featuring larger context lengths, higher memory, and performance/TCO

Prefill optimized

High FLOP/W, compute-bound workloads

Software Features

Broad model support

Supports language, reasoning, multimodal, and diffusion models

KV cache efficiency

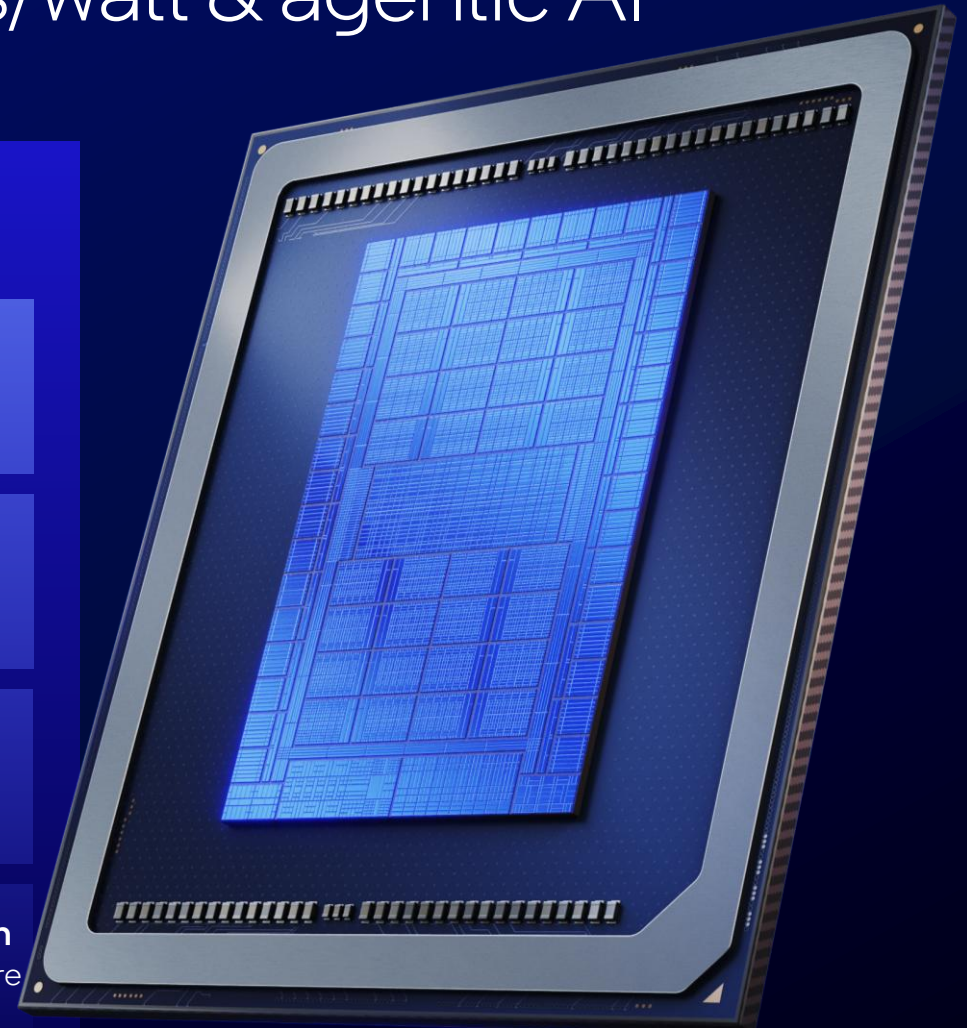
Enables KV-cache-aware routing and offloading for scalable inference

Open framework ecosystem

Built to work with leading AI serving frameworks

Heterogeneous agent orchestration

Runs across heterogeneous infrastructure without changing agent code



An Industry Standard AI Development Stack

Open, Upstreamed and Day 0
Ready for Intel® GPU

Trusted by an Ecosystem

Used by 100s of ISVs across a wide variety of applications

Built to Feel Familiar

Open ecosystems, and developer-trusted libraries & frameworks

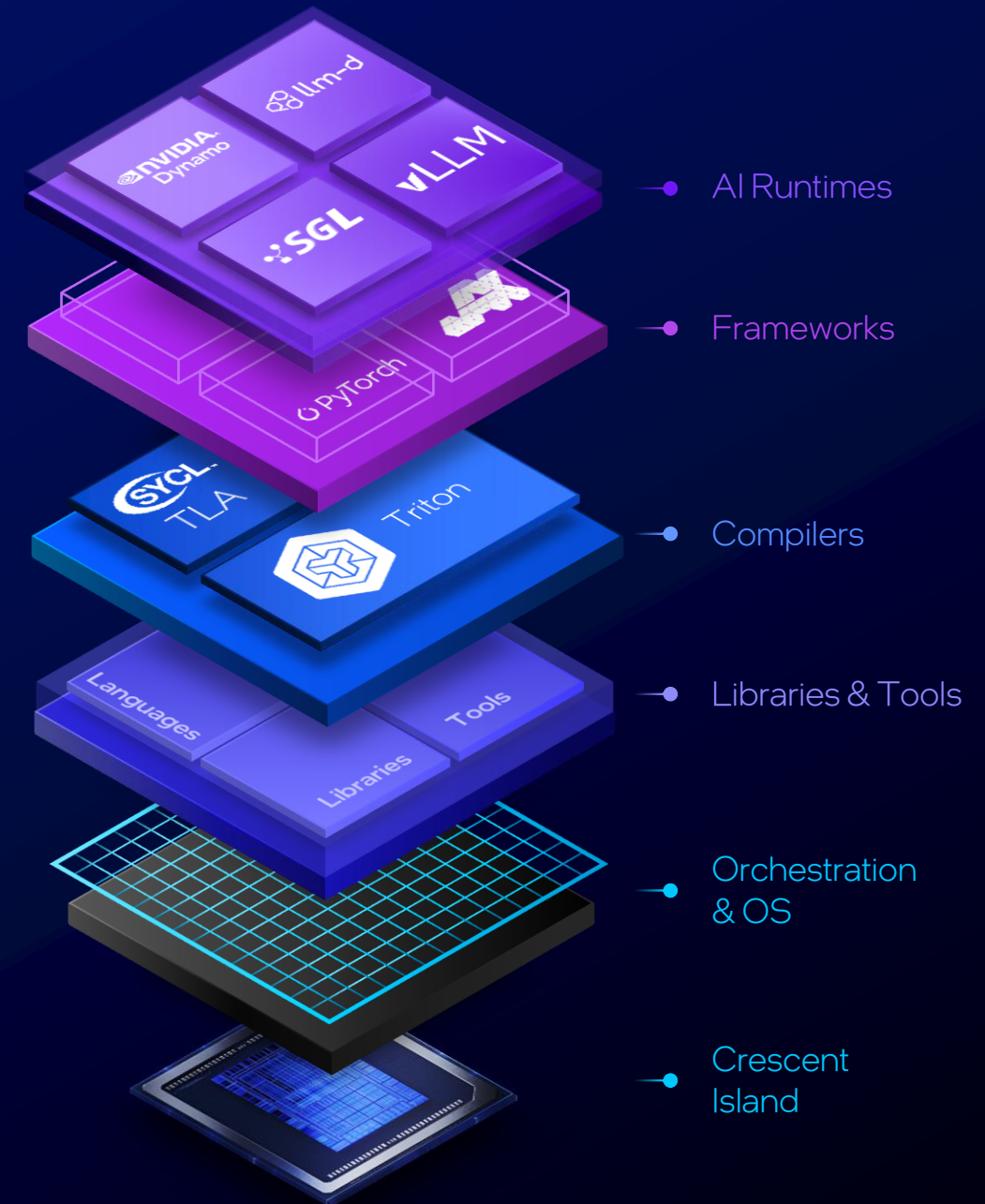
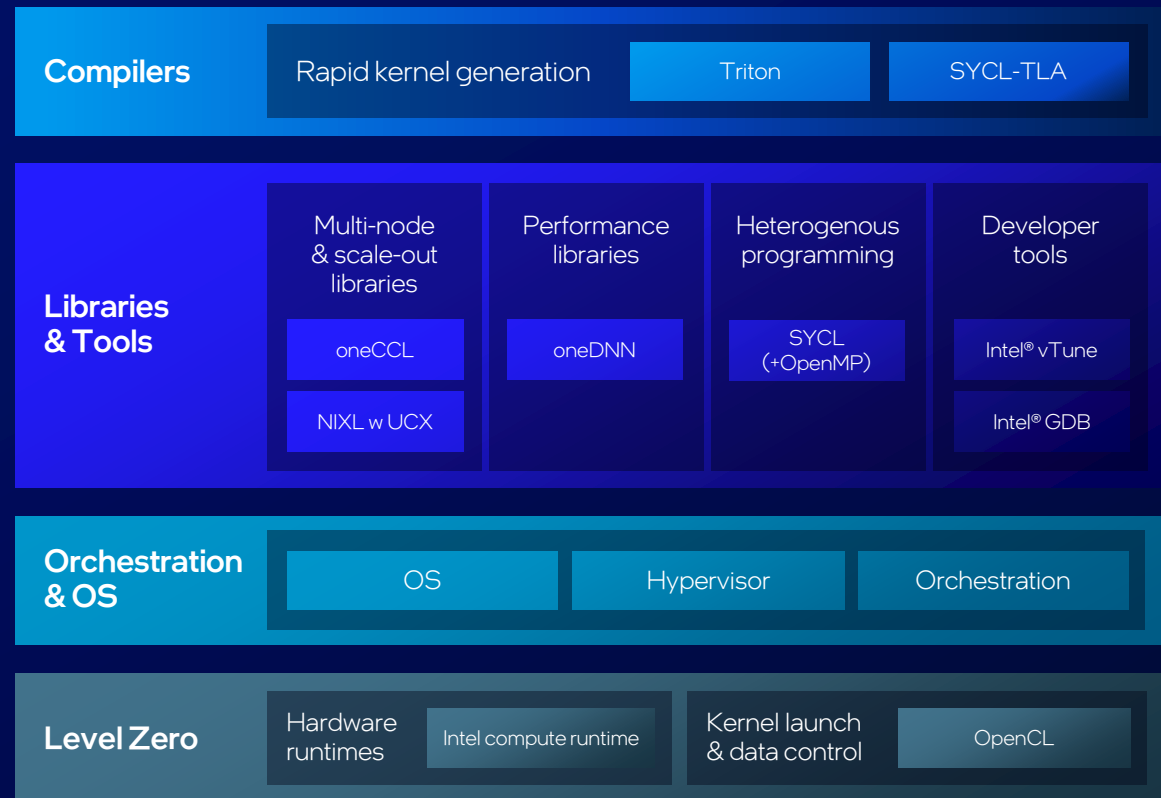
Support You Can Rely on

Day 0 functionality and performance with fully validated firmware, drivers and reference systems



An Industry Standard AI Software Stack

Open, Upstreamed and Day 0
Ready for Intel® GPU



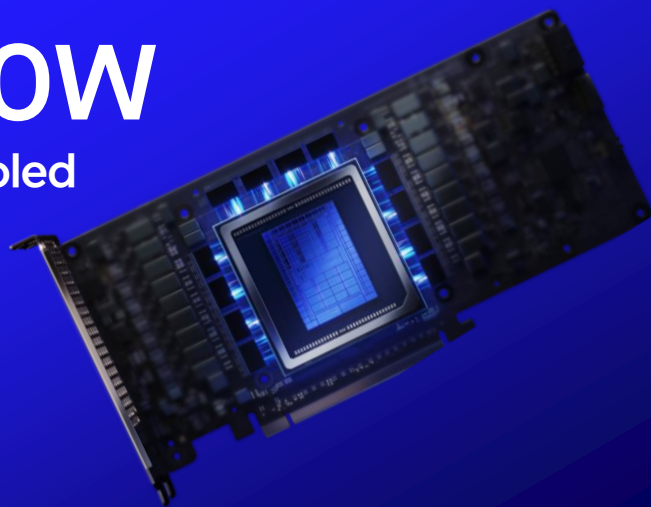
Intel® GPU

Code-named
Crescent Island

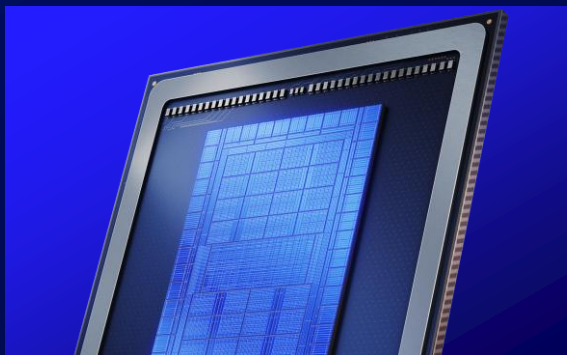
Performant. Efficient.
Inference optimized GPU for agentic AI.

350W

Air cooled
PCIe



Up to **480 GB** **LPDDR 5x** Memory



Agentic AI Ready

more concurrent sessions, long context workloads, and larger models

Widest range of Data types
from FP4-FP64

Prefill optimized

High FLOP/W, optimized for compute-bound workloads

Open
software stack

Industry
standard
frameworks
Day 0
functionality



32 X^e Cores

X^e3P GPU IP

intel®

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Methodology & Sources: Bytes Held Grew 7x, Bytes Read Per Token Fell 4x

Methodology

- Weight footprint only, on a linear axis. Candidate replacement, not an addition.
- Model configurations were read from config.json in each vendor's public Hugging Face repository, `huggingface.co/<repo id>/blob/main/config.json`
- arXiv references correspond to the model publications listed. arXiv ids resolve at `arxiv.org/abs/<id>`.
- DeepSeek-V4-Flash, DeepSeek-V4-Pro and Kimi K3 2.8T are announced only, configurations not yet published; drawn with dashed rings and carrying no headline claim.
- Weights charged at each checkpoint's native format. KV cache excluded: this is the weight term only.

Models & Data Sources

- Llama 2 70B: meta-llama/Llama-2-70b-hf, arXiv:2307.09288.
- Mixtral 8x22B: mistralai/Mixtral-8x22B-v0.1, arXiv:2401.04088.
- DeepSeek-V2: deepseek-ai/DeepSeek-V2, arXiv:2405.04434.
- Llama 3.1 405B: meta-llama/Llama-3.1-405B, arXiv:2407.21783.
- DeepSeek-V3 / R1: deepseek-ai/DeepSeek-V3, arXiv:2412.19437 and arXiv:2501.12948.
- Llama 4 Maverick: meta-llama/Llama-4-Maverick-17B-128E-Instruct (gated; config read from mirror unsloth/Llama-4-Maverick-17B-128E-Instruct), Meta AI blog 5 April 2025.
- Qwen3-235B-A22B: Qwen/Qwen3-235B-A22B, arXiv:2505.09388.
- gpt-oss-120b: openai/gpt-oss-120b, arXiv:2508.10925.
- Kimi K2 1T: moonshotai/Kimi-K2-Instruct, arXiv:2507.20534.
- Qwen3-Next-80B-A3B: Qwen/Qwen3-Next-80B-A3B-Instruct, Qwen Team blog 10 September 2025.
- Kimi Linear 48B-A3B: moonshotai/Kimi-Linear-48B-A3B-Instruct, arXiv:2510.26692.

Methodology & Sources: Decode Becomes a Compute Problem

Methodology

- Accepted lengths from the LMSYS SpecBundle performance dashboard.
- Drafting method EAGLE-3, Li, Wei, Zhang and Zhang, arXiv:2503.01840, served by SGLang at batch 8. Means over `gsm8k`, `math500`, `mtbench`, `humaneval`, `livecodebench`, `financeqa`, `gpqa`.
- Expert counts from each model's `config.json`.
- The source also publishes wall-clock throughput; it is deliberately not shown.
- Full record: `Speculative_Decoding_Slide_References.md`.

Models & Data Sources

- SpecForge project (LMSYS Org / SGLang)
 - `docs.sglang.io/SpecForge/SpecBundle`.
 - Data file: `sgl-project.github.io/SpecForge/SpecBundle/raw_data/data.json`, retrieved 13 August 2026,
 - archived verbatim as `specbundle_raw_data.json`.
 - Repository: `github.com/sgl-project/SpecForge`, Apache License 2.0.
- Announcements:
 - `lmsys.org/blog/2025-07-25-spec-forge/`
 - `lmsys.org/blog/2025-12-23-spec-bundle-phase-1/`.
- Targets as named in the dashboard:
 - Qwen3-30B-A3B-Instruct-2507, Qwen3-235B-A22B-Instruct-2507, Qwen3-Next-80B-A3B-Instruct-FP8, Qwen3-Coder-30B-A3B-Instruct, Qwen3-Coder-480B-A35B-Instruct, Kimi-K2-Instruct, Ling-flash-2.0, Llama-4-Scout-17B-16E-Instruct, Llama-3.3-70B-Instruct.

The Intel logo is centered on a dark blue background. It features a small, bright blue square positioned above the first vertical stroke of the lowercase letter 'i'. The word 'intel' is rendered in a clean, white, sans-serif typeface. A registered trademark symbol (®) is located at the end of the word, to the right of the final period.

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